VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

REG. NO.:

SCHOOL OF ADVANCED SCIENCES
DEPARTMENT OF MATHEMATICS
CONTINUOUS ASSESSMENT TEST – II
FALL SEMESTER – 2025~2026

SLOT: C1+TC1+TCC1

Programme Name & Branch	:	B.Tech. (All Branches)
Course Code & Course Name	:	BMAT205L & Discrete Mathematics & Graph Theory
Faculty Name(s) & Class Number(s)	:	Common Question Paper
Date of Examination and Session	:	07-10-2025 & AN Session
Exam Duration	:	90 Minutes Maximum Marks : 50 M

General Instruction(s):

- Answer All Questions.
- M – Max. Marks; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create).
- Course Outcomes:
  CO-2: Use algebraic structures in applications, CO-3: Counting techniques in engineering problems, CO-4: Use lattice and Boolean algebra properties in Digital circuits.
- Students are permitted to carry any number of text books and hand written class note books.

Q. No.	Questions	M	CO	BL
1.	Find the code words generated by the parity check matrix $H = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$, when the encoding function is $e : B^3 \longrightarrow B^6$. Also find how many errors e will detect? and how many errors it will correct?	10	2	3
2.	A database assigns unique **record IDs** numbered from 1 to 5000. For system optimization, IDs that are divisible by 3, 5, or 7 are reserved for internal indexing and cannot be given to user records. How many valid record IDs can be assigned to users for internal indexing?	10	3	2

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Q. No.	Questions	M	CO	BL
3.	(i). How many different 3 –digit numbers can be formed using the digits 1, 2, 3, 4, 5 without repetition? (ii). In how many ways can the letters of the word **MATH** be arranged? (iii). From 7 men, a committee of 3 is to be formed. In how many ways can this be done? (iv). How many different 4 – letter words can be formed using the letters of the word **BOOK**? (v). In how many ways can 5 different books be arranged on a shelf?	10	3	1
4.	A city decides to plant trees along a highway. In the first year, 50 trees are planted. In the second year, 120 trees are planted. From the third year onwards, the number of trees planted in a year is equal to the sum of trees planted in the previous year and twice the number of trees planted two years earlier. 1. Formulate a recurrence relation for the number of trees planted in year n. 2. Find the number of trees planted in the nth year. 3. Compute the total number of trees planted by the end of the 4th year.	10	3	3
5.	(1). Let $X_1 = \{S_0, S_1, S_2, \cdots, S_7\}$ where $S_0 = \{a, b, c, d, e, f\}$, $S_1 = \{a, b, c, d, e\}$, $S_2 = \{a, b, c, e, f\}$, $S_3 = \{a, b, c, e\}$, $S_4 = \{a, b, c\}$, $S_5 = \{a, b\}$, $S_6 = \{a, c\}$ and $S_7 = \{a\}$. (i). Verify whether $(X_1, \subseteq)$ is a partially ordered set or not. (ii). If so, draw a Hasse diagram for the same. (2). Consider the set $X_2 = \{2, 3, 6, 12, 24, 36\}$ and let for $a, b \in X_2$, "$a \leq b$" be the relation "a divides b" (i). Verify whether X_2 is a Poset or not. (ii). Draw the Hasse diagram of $(X_2, \leq)$. (iii). Check $(X_2, \leq)$ is lattice or not.	10	4	3

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